

Loan Default Risk Analysis: Underwriting Strategy for a \$32.58B Portfolio

Executive Summary

This project analyzes a \$32.58B loan portfolio from the perspective of a Chief Risk Officer (CRO) tasked with stopping capital erosion. The portfolio is currently generating a **-15.33% net profit margin**, translating to a **\$569.77M net financial loss**, driven by \$4.29B in defaulted loans. Using Python for data cleaning and Power BI for modeling and visualization, this analysis identifies the specific borrower profiles, loan structures, driving default risk and where underwriting standards should be tightened to stop further losses.

Business Problem

Although the overall default rate is 11.61%, the risk is not evenly distributed across all borrowers. This project identifies the customer groups and loan characteristics that contribute most to defaults, helping the business improve its lending strategy while continuing to approve low-risk borrowers.

Data & Methodology

- **Source:** Kaggle
- **Pipeline:** Python (data cleaning) → Power BI (modeling, DAX, visualization)

Modeling Decisions

- **Continuous variables** such as **Debt-to-Income (DTI)**, **Loan-to-Income (LTI)**, **Interest Rate**, and **Income** were grouped into meaningful ranges (e.g., **Under \$30K**, **21–40% DTI**, **21–25% Interest Rate**). This made it easier to compare borrower groups and identify high-risk segments.
- A **custom DAX measure** was created to automatically highlight the highest default-risk category in each chart. The highlighted category updates dynamically whenever users apply filters or slicers, making it easier to identify the highest-risk borrower group.

Limitations

- The dataset did not include **loan recovery information**, so financial losses are based on the total defaulted loan amount rather than the actual amount recovered.
- **Gender data** was not available, so default risk could not be analyzed by gender.
- The dataset did not include **year or date information**, so it was not possible to analyze trends over time. The results represent a single snapshot of the loan portfolio.

Key Findings

1. Portfolio Performance

The loan portfolio is currently operating at a loss. Although it generates \$3.72B in expected interest income, \$4.29B in loan defaults resulted in a \$569.77M net financial loss and a -15.33% profit margin. In addition, 75.11% of borrowers fall into high-risk categories, and the portfolio has a high-risk concentration (HHI score: 3.23K).

2. Low-Income Borrowers Have the Highest Risk

Borrowers earning less than \$30K per year have the highest default rate (21.96%), which is much higher than the \$30K–\$50K income group (13.55%). This suggests that low-income applicants may require additional review before loan approval.

3. High Loan Burden Increases Default Risk

Borrowers who default have a much higher Loan-to-Income (LTI) ratio (325%) compared to the portfolio average (218%). This indicates that borrowers with very high loan amounts relative to their income are more likely to default.

4. Certain Loan Types Carry Higher Risk

Some loan characteristics have higher default rates than others:

- Business loans have the highest default rate (12.33%) among all loan purposes.
- Loans with 21–25% interest rates have the highest default rate (17.82%).
- Loans with a 37–48% month term have the highest default rate (13.51%).

5. Borrower Demographics Affect Default Risk

Some borrower groups are more likely to default than others:

- Borrowers aged 18–24 have the highest default rate (21.03%).
- Unemployed borrowers have the highest default rate (29.18%).

- Borrowers with a High School education have the highest default rate among education groups (12.88%).
- Divorced borrowers have the highest default rate (12.53%) among marital status groups.

Dashboard Overview

The Power BI dashboard is built across three pages:

- **Overview:** Portfolio-level KPIs portfolio value, expected income, net financial result, profit margin, client-at-risk rate, default rate, DTI/LTI averages, and a summarized key-insights panel.
- **Demographics Analysis:** Default risk broken down by age, employment status, education, marital status, and dependents/co-signer status.
- **Loan Analysis:** Default rate broken down by DTI category, LTI category, credit tier, loan purpose, income tier, loan term, and interest rate bracket.

Each page uses red highlighting (via the custom DAX measure) to dynamically call out the highest-risk segment as the user filters the report.

Recommendations for Underwriting

Based on the analysis, the company should consider the following:

- Apply stricter approval checks for applicants earning less than \$30,000.
- Set a lower Loan-to-Income (LTI) limit to reduce the risk of lending to overleveraged borrowers.
- Review business-purpose loans more carefully before approval, as they have the highest default rate.
- Pay extra attention to loans with 37–48 month terms and 21–25% interest rates, as these show higher default risk.
- Identify applicants with multiple risk factors (such as young age, unemployment, and low income) and conduct additional credit reviews before approving the loan.

Limitations & Next Steps

This analysis is a cross-sectional snapshot without a year table, no trend or seasonality view was possible, so it's unclear whether these risk patterns are worsening, stable, or improving over time. Without a loss/severity table, the \$4.29B default figure reflects gross exposure rather than net-of-recovery loss, likely overstating the true economic loss. With more data access, next steps would include:

- Adding a time dimension to track whether risk concentration is trending up or down
- Incorporating recovery/severity data to model net loss more precisely
- A demographic fairness review (gender data) to ensure tightened criteria don't introduce disparate impact

Skills Demonstrated

Python (data cleaning), Power BI (data modeling, DAX), advanced DAX (MAXX, ALLSELECTED, dynamic content filtering), data binning/discretization for executive reporting, dashboard UX design, risk analysis and business framing.